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Abomasal infusions of essential and nonessential amino acids to evaluate productive efficiencies, metabolism, energy, and amino acid utilization in lactating dairy cattle

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ABSTRACT

The use of MP in diet formulation incorporates both EAA and nonessential amino acids (NEAA). Further, under conditions of high metabolic demand, NEAA synthesis might be energetically unfavorable and could be a limiting step if other substrates are limited. The objective of this study was to evaluate production outcomes and metabolism of cows fed a diet meeting the ME requirement but MP-limited with abomasal infusions of NEAA and EAA. Twelve multiparous Holstein cows were randomly assigned to 1 of 5 treatments in a replicated 5 × 5 Latin square crossover design: (1) water to meet 90% of EAA and NEAA (90AA; 0 g/d); (2) EAA to meet 100% of EAA and 90% of NEAA (100EAA; 188 g/d); (3) 90% of EAA and 100% of NEAA (100NEAA; 158 g/d); (4) 100% of EAA and NEAA (100AA; 346 g/d); or (5) 100% of EAA and 110% of NEAA (110NEAA; 504 g/d). Continuous infusions were conducted for 18 d, and sampling occurred on d 15 to 18 for milk, blood, diet, refusals, and feces. Dry matter intake and milk yield were recorded daily. Data were analyzed by fitting a mixed-effects model and Tukey pairwise comparison using R (version 4.2.0; R Core Team). Body weight, DMI, milk yield, protein, and lactose were not different among treatments. Milk fat yield and concentration trended lower and milk total fatty acid (FA) content was lowest in the 110NEAA treatment. Preformed FA in milk were lower for the 110NEAA treatment compared with the 100EAA, whereas the opposite was observed for mixed FA. Plasma insulin concentrations trended to be lower for the 100EAA and 100NEAA compared with the 110NEAA treatments. Although total plasma AA did not differ among treatments, plasma NEAA and EAA responded to the respective infusions. Overall, infusing NEAA above the MP requirement tended to increase insulin, which is related to lower lipolysis, and this was supported by a significant decrease in preformed milk FA yield and concentration.

Key words: amino acid, abomasal infusion, energy, milk fatty acid

INTRODUCTION

The predicted MP supply is the summation of EAA and nonessential amino acids (**NEAA**). The Cornell Net Carbohydrate and Protein System (**CNCPS**) nutritional model (Higgs, 2014; Higgs et al., 2023) predicts supplies of 9 individual EAA supplies and 1 conditionally EAA (Arg), but no such estimation of NEAA supply. Most studies evaluating AA supply in dairy cattle diets have focused mainly on EAA formulation, with little consideration for the supply of individual NEAA. However, in addition to being building blocks of proteins, NEAA play extensive metabolic roles throughout the body and are not entirely dispensable AA. A study from Metcalf et al. (1996), where the infusion of a mixture of only EAA yielded a milk protein yield response as well as an infusion of a mixture of total AA (**TAA**), suggested that NEAA might have a limited effect on milk protein yield response. Furthermore, there is a lack of data to determine whether minimal dietary requirements exist for nutritionally relevant NEAA (Wu et al., 2013, 2014). The critical functions of NEAA have been reiterated for metabolism, lactation, growth, reproduction, and health (Wu, 2009, 2010, 2021), mainly in monogastric animals, with limited data in lactating dairy cattle. The concept of NEAA requirements could seem counterintuitive, given that they have traditionally been considered not required for diet formulation due to the animals' capability to synthesize NEAA in appropriate levels de novo (Wu et al., 2013). However, de novo synthesis of NEAA occurs from the

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

C and N of several substrates, including the EAA, which are taken up in excess by the mammary gland (**MG**; Doepel and Lapierre, 2010) and in other tissues for other purposes (Wu, 2014). To maintain whole-body homeostasis, all AA are required to interact with each other and other metabolites across all tissues to regulate metabolic pathways, gene expression, immunity, oxidative defense, secretagogues, protein turnover, and cell signaling and physiology (Wu, 2009). Therefore, even though the body can produce NEAA from other substrates, they are metabolically required, and their synthesis can be energetically demanding. If other substrate resources are limited, the synthesis and availability NEAA might be a limiting step.

There have been numerous infusion studies of EAA, either of all EAA, mixtures of EAA, and the individual EAA, particularly of the most limiting EAA in lactating ruminant diets: Met, Lys, and His (e.g., Schwab et al., 1976; Appuhamy et al., 2011; Zanton et al., 2014). Moreover, because casein makes up most of the protein in milk, it has been used in infusion studies on lactating dairy cows and transition cows, with modest milk protein or milk yield response (Schwab et al., 1976; Choung and Chamberlain, 1993; Larsen et al., 2014). Earlier studies in ruminants had some focus on a few NEAA (Mepham and Linzell, 1966; Schwab et al., 1976) but mainly regarding glucose catabolism and metabolic fate of AA (Black et al., 1955, 1968; Wolff and Bergman, 1972). Most of the more quantitative data on NEAA metabolism have been generated in studies with monogastric animals (Wu, 2009; Wu et al., 2013, 2014). In lactating cows, the AA profile of casein was used to evaluate production responses of EAA and NEAA but also with minimal response (Metcalf et al., 1996; Doepel and Lapierre, 2010). In the current work, we have the capacity to formulate diets with much more comprehensive information than has been done previously (Higgs, 2014; LaPierre, 2021). Indeed, the application of CNCPS v7 (Higgs et al., 2023) allows diets to be formulated to target a more precise EAA supply, which, by calculating the difference, should provide a more quantitative estimate of the NEAA supply and requirements on a macro scale. Thus, the objectives for this study were to investigate whether and how productive efficiency of a lactating cow can be modified when both EAA and NEAA requirements are deficient or met, and whether excess NEAA supply would result in an increase or decrease in the efficiency of both energy and AA. Further, the approach of using ME as the basis of estimating AA requirements will provide an updated procedure to quantitatively determine whether NEAA are critical AA requirements of the lactating dairy cow. The hypothesis is that supplying or limiting both EAA and NEAA alters productive and metabolic responses by allocating substrates usually used for de novo synthesis of NEAA toward other purposes.

MATERIALS AND METHODS

Experimental Design and Diets

The Cornell University Institutional Animal Care and Use Committee approved all animal procedures (protocol 2021-0064). Sixteen Holstein cows were surgically fitted with rumen cannulas at the start of their dry period at the Cornell University Ruminant Center (Harford, NY). Twelve of these cows in their second and third lactations (n = 10 and n = 2, respectively) were blocked by DIM into 2 blocks. At the beginning of the covariate period, cows in block 1 averaged 75 DIM ± 27 d and block 2 averaged 66 DIM ± 13 d. Each cow was randomly assigned a unique treatment sequence in a replicated 5 × 5 Latin square crossover design (5 periods and 5 treatments) with 18-d periods implemented in 2 blocks of 6 cows. Each cow received all 5 treatments, a different treatment each period, and every cow had a unique sequence of treatments to mitigate potential residual carryover effects. Animals were housed in individual metabolism stalls, fed a TMR once daily at 0630 h, and milked 3 times per day at 0600, 1400, and 2200 h. The study began with an 18-d covariate period, during which all cows were fed the same diet targeting 100% of MP and ME requirements (**Cov Diet**), with samples collected on d 16 to 18. Requirements were calculated using CNCPS (v.6.5.5 and v.7; Van Amburgh et al., 2015; Higgs and Van Amburgh, 2016; Higgs et al., 2023) and based on second-lactation animals at 41 mo of age, BW 690 kg (mature BW 800 kg), and milk yield 45 kg/d with 3.35% protein, 4.60% fat, and 4.63% lactose. For 3 d after the conclusion of all periods, cows were fed the Cov Diet, and no infusions were performed, as a short washout. Infusion lines were placed into the abomasum through the rumen cannula after the covariate period, and lines were checked 3 times per week to ensure that infusion devices remained in the abomasum. Further details on the infusions are described in the next section. During the 5 experimental periods, all cows were fed a similar diet (**Exp Diet**) while simultaneously receiving their respective infusion treatment. The Exp Diet was formulated to provide 100% of ME requirements and 90% of EAA and NEAA requirements via MP balance using CNCPS (v.6.5.5 and v.7; Van Amburgh et al., 2015; Higgs and Van Amburgh, 2016; Higgs et al., 2023). It was assumed that if both MP and EAA supplies were formulated for 90% of animal requirements, then NEAA supply would be 90% of requirements. The ingredient compositions of the Cov Diet and the Exp Diet are summarized in Table 1. Due to a forage management situation, there was a change from conventional corn silage (**CS**) to brown-midrib corn silage (**BMR**) and grass silage bunks after the first experimental period; therefore, 2 Exp Diets are described in this manuscript.

Table 1. Ingredient inclusion (% DM ± SD) in the covariate (Cov Diet) and experimental diets (Exp Diet)[1]

Item	Cov Diet (Covariate period)	Exp Diet 1[2] (Period 1)	Exp Diet 2[2] (Periods 2–5)
Corn silage (CS)/brown midrib CS	40.21	39.90	40.48 ± 0.39
Grass silage bunk 1/grass silage bunk 2	19.17	17.78	17.84 ± 0.25
Corn meal	13.09	14.59	14.38 ± 0.19
Heat-treated soybean meal[3]	6.10	2.54	2.51 ± 0.04
Citrus pulp	4.18	4.89	4.82 ± 0.07
Wheat middlings	4.18	5.34	5.26 ± 0.07
Dextrose	4.07	4.61	4.55 ± 0.06
Soybean meal	2.03	3.49	3.44 ± 0.05
Blood meal	1.83	1.27	1.26 ± 0.02
Fat supplement[4]	1.01	1.07	1.05 ± 0.02
Palmitic acid supplement[5]	1.01	1.07	1.06 ± 0.02
Urea 281	0.23	0.46	0.46 ± 0.01
Rumen-protected methionine[6]	0.10	0.06	0.07 ± 0.01
Rumen-protected AA[6]	0.11	0.08	0.09 ± 0.01
Live yeast[7]	0.03	0.04	0.04 ± 0.01
Monensin sodium[8]	0.01	0.01	0.01 ± 0.01
Vitamins and minerals	2.64	2.81	2.77 ± 0.04
Sodium bicarbonate	0.74	0.79	0.78 ± 0.02
Calcium magnesium carbonate[9]	0.54	0.57	0.57 ± 0.01
Limestone, ground	0.42	0.41	0.41 ± 0.01
Dicalcium phosphate	0.41	0.46	0.46 ± 0.01
Salt, white	0.30	0.32	0.32 ± 0.01
Vitamin–trace mineral premix[10]	0.20	0.22	0.22 ± 0.01

[1]All cows were fed the same diet during the corresponding period, regardless of treatment.

[2]The same grain mix was used for both experimental diets; forages in Exp Diet 1 were corn silage and grass silage 1, and forages in Exp Diet 2 were brown midrib CS and grass silage 2.

[3]AminoPlus, Ag Processing Inc., Omaha, NE.

[4]Energy Booster 100, Milk Specialties, Eden Prairie, MN.

[5]Palmit 80, Global Agri Trade Corporation, Rancho Dominguez, CA.

[6]Smartamine M and Smartamine ML, Adisseo USA Inc., Alpharetta, GA.

[7]Levucell SC, Lallemand Inc., Milwaukee, WI.

[8]Rumensin, Elanco Animal Health, Greenfield, IN.

[9]MIN-AD, Papillon, Eason, MD.

[10]PAN Dairy VTM: Ca, 27% DM; Fe 762 mg/kg; Zn 26,168 mg/kg; Cu 5,947 mg/kg; Mn, 19,073 mg/kg, Se, 132 mg/kg; Co, 587 mg/kg, I, 587 mg/kg; vitamin A, 2,978 KIU/kg; vitamin D, 580 KIU/kg; vitamin E, 14,480 KIU/kg.

Treatments and Abomasal Infusion Setup

The 5 treatments were as follows: (1) water only, as a control, to meet 90% requirement of EAA and NEAA (**90AA**); (2) EAA mix at 184.8 g/d to meet 100% requirement of EAA and 90% requirement of NEAA (**100EAA**); (3) NEAA mix at 158.0 g/d to meet 90% requirement of EAA and 100% requirement of NEAA (**100NEAA**); (4) EAA and NEAA mix at 344.1 g/d to meet 100% requirement of EAA and NEAA (**100AA**); and (5) EAA and double NEAA mix at 501.9 g/d to meet 100% requirement of EAA and 110% requirement of NEAA (**110NEAA**).

The following steps were taken to calculate the individual AA quantity to be infused. First, a diet was formulated using the CNCPS (v.6.5.5 and v.7) to meet 100% ME and MP requirements, as in the Cov Diet. This diet also met optimal grams of EAA per megacalorie of ME, based on the recommendations proposed by Higgs et al. (2023). Then, a second diet was formulated to provide 90% of the MP supply while maintaining 100% ME requirements. Next, the grams of MP supply from the second diet were subtracted from the first diet, to calculate the TAA that needed to be infused to meet 100% of MP requirements. The same subtraction was performed for each EAA, as the model provides this information. For NEAA amounts, a profile (percent of total NEAA) was determined for microbial and dietary protein using data from our laboratory, generated through multiple time-point hydrolysis (Ortega et al., 2025). The AA Glu and Gln were analyzed as one (**Glx**), as were Asp and Asn (**Asx**), due to the analytical procedures used to determine the NEAA profile. By subtracting the grams of MP of EAA from TAA, the total grams of NEAA that needed to be infused was calculated to meet 100% of MP NEAA perceived requirements. Finally, from the determined profile, the percentage of each NEAA (percent total NEAA) was multiplied by the total grams of NEAA to calculate the individual NEAA amount that needed to be infused. To address solubility issues, 75% of Tyr was replaced with Phe, whereas Glx was partitioned as 33% Gln and 67% Glu, and Asx was divided as 25% Asn and 75% Asp. The individual AA amounts infused in each treatment are detailed in Table 2.

Table 2. Daily amounts of EAA and nonessential amino acids (NEAA) infused into the abomasum of cows per treatment

	Treatment				
AA, g/d	90AA	100EAA	100NEAA	100AA	110NEAA
Arg	—	18.4	—	18.3	18.3
His	—	16.0	—	15.9	15.9
Ile	—	9.3	—	9.3	9.3
Leu	—	36.7	—	36.5	36.5
Lys	—	26.8	—	26.7	26.7
Met	—	15.7	—	14.5	14.9
Phe[1]	—	21.5	9.0	30.7	40.0
Thr	—	15.0	—	14.9	14.9
Trp	—	4.3	—	4.9	4.9
Val	—	21.1	—	24.0	24.0
Total EAA	0.0	184.8	9.0	195.6	205.4
Ala	—	—	20.7	20.7	41.4
Asx[2]	—	—	33.6	33.6	67.2
Cys	—	—	3.9	3.6	8.0
Glx[3]	—	—	38.7	38.7	77.4
Gly	—	—	18.0	18.0	35.9
Pro	—	—	16.0	16.0	32.0
Ser	—	—	15.2	15.2	30.4
Tyr[1]	—	—	3.0	2.8	6.2
Total NEAA	0.0	0.0	149.2	148.6	298.6
Total AA	0.0	184.8	158.2	344.1	503.9

[1]Due to solubility issues, 75% of Tyr was replaced with Phe.

[2]Due to solubility issues, Asx was partitioned as 25% Asn and 75% Asp.

[3]Due to solubility issues, Glx was partitioned as 33% Gln and 67% Glu.

All treatments were infused continuously, except when cows went to the milking parlor, using Masterflex LS Standard Digital Drive Pumps installed with L/S Easy-Load 3 peristaltic pump heads (Cole Parmer, Vernon Hills, IL). A platinum-cured silicone Masterflex L/S Precision Pump tubing (L/S 17, 6.4 mm i.d. × 9.5 mm o.d.) was used from the carboys to the peristaltic pump head and connected to a flexible hybrid PVC tubing (9.5 mm i.d. × 12.7 mm o.d.). The latter tubing was connected to a valve (Figure 1A, 1B) to allow the cows to be disconnected for milking without losing any AA mixture. That same tubing was then attached to a modified version of the stainless-steel infusion device described by Westreicher-Kristen and Susenbeth (2017). Filters with 100-µm pore size were placed around the device to prevent clogging from gastrointestinal contents, and a flange was made from light-duty multi-ply conveyor belt material (4.8-mm thick) to keep the device in the abomasum (Figure 1C). All AA mixtures were prepared in Nalgene bottles using hot water 2 d before delivery to the farm. The 100EAA and 100NEAA mixtures were prepared every 4 d and infused at 258 mL/h. The 100AA mixture was prepared every 3 d, and the 110NEAA was prepared every 2 d, and both were infused at 510 mL/h. Water for the 90AA treatment was replaced as needed and infused at both 258 mL/h and 510 mL/h, based on the treatment that the cow in the closest stall was receiving.

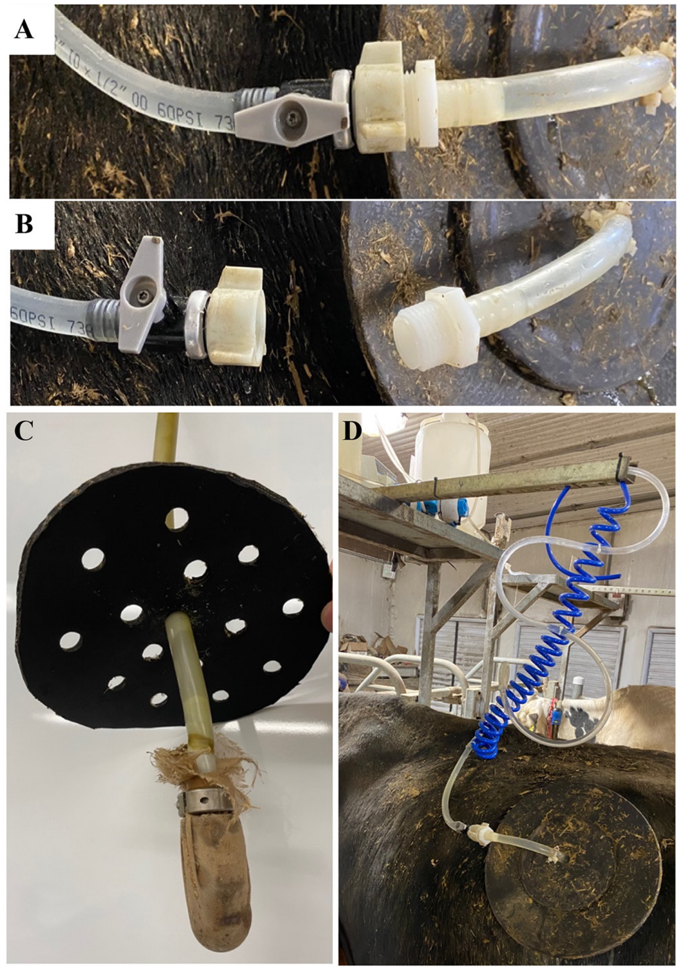

Figure 1. (A) Open and attached connection valve. (B) Closed and unattached connection valve, when cows went to the parlor. (C) Modified infusion device with filter and flange. (D) On-farm setup of the infusion line with coiled tubing, peristaltic pump, and carboys with AA mixtures.

Sampling and Analysis

Animal Measurements and Intake. Body weights and BCS (1–5 scale) were measured and recorded on d 1, 5, 10, 14, and 18 of each period. Body condition scoring was conducted by 2 trained scorers. Both DMI and refusals for each cow were measured daily throughout the experiment.

Diet, Feeds, and Feces. During d 1 to 14 of each period, TMR was sampled 3 times per week and composited for analysis. During sampling from d 15 to 18, samples of TMR and forages were collected daily and composited for each period. Refusals were collected daily during sampling and composited for each cow. Premixes were manufactured at a commercial feed mill (Purina Animal Nutrition, Caledonia, NY) and delivered to the farm. As a new batch of premix was delivered to the farm, a subsample was collected for analysis. Similarly, as each batch arrived, individual ingredients used in the premix of corn grain, citrus pulp, soybean meal (**SBM**), heat-treated SBM, wheat middlings, and bloodmeal were analyzed for chemical composition. The TMR, refusals, and forages were analyzed for nutrient composition using near-infrared reflectance spectroscopy (NIR 5 with plus option), and the concentrate mixture and ingredients were analyzed using wet chemistry (CNCPS package) by Cumberland Valley Analytical Services (Waynesboro, PA). Subsamples of the composite TMR, refusals, and forages, as well as the individual premix ingredients, were dried in a forced-air oven at 60°C and ground to 1 mm using a Wiley mill for further nutrient analysis. The TMR and refusal samples were analyzed for DM, sodium sulfite and alpha amylase–treated NDF corrected for organic matter (**aNDFom**), and in vitro sodium sulfite and alpha amylase–treated NDF after 240 h in-vitro rumen fermentation corrected for organic matter (**uNDF240om**) as described by Raffrenato et al. (2018). Forages and premix ingredients were analyzed for AA using zwitterionic hydrophilic interaction liquid chromatography (**Z-HILIC**) and triple-quadruple liquid chromatography–mass spectrometry (**LCMS**) as described by Ortega et al. (2024). Briefly, 3 methods were used to analyze 18 AA: (1) acid hydrolysis using 0.1% phenol 6 *M* HCl; (2) performic acid oxidation before acid hydrolysis for sulfur

AA; and (3) alkaline hydrolysis using barium hydroxide for Trp. All methods hydrolyzed samples for 24 h at 110°C in an oxygen-free environment, and supernatants were filtered through a 0.2-µm syringe filter and diluted to appropriate concentrations depending on the N content. The nutrient composition of all diet ingredients analyzed was input into CNCPS (v.6.5.5) for the prediction of ME, MP, and AA supplies of the diet. Fecal sampling for each cow occurred on d 15 to 17 of every period (d 15: 1100 h, 1700 h, 2300 h; d 16: 0500 h, 1400 h, 2000 h; d 17: 0200 h, 0800 h). The 8 samples collected for each cow were composited, dried in a forced-air oven, and ground to 1 mm. Fecal samples were also analyzed for DM, aNDFom, and in vitro uNDF240om. Results from the TMR, refusals, and feces were used to estimate the diet's apparent total-tract aNDFom digestibility using uNDF240om as an internal marker (Huhtanen et al., 1994; Raffrenato et al., 2018).

Milk. Cows were milked 3 times per day, and milk weights were recorded at each milking using DelPro FarmManager software (DeLaval, Kansas City, MO). On d 15 to 18 of each period, milk samples were collected at every milking, and composition analysis was performed in the Barbano Laboratory at the Department of Food Science at Cornell University (Ithaca, NY) for true protein, fat, anhydrous lactose, MUN, and fatty acids (**FA**) using a Fourier-transform infrared (**FTIR**) spectrophotometer (Lactoscope model FTA, Delta Instruments, Drachten, the Netherlands). Milk fat, true protein, anhydrous lactose, and MUN were calibrated on the FTIR using a modified milk calibration sample set with reference chemistry values as described by Kaylegian et al. (2006) and Portnoy et al. (2021). Fatty acid sources were analyzed via FTIR using predicted least squares prediction models described by Woolpert et al. (2016). Calibration was carried out using gas–liquid chromatography reference chemistry as described by Wojciechowski and Barbano (2016). The calibration sets used for milk components and FA analysis were as follows: 0.05 to 1.40 g/100 g milk for de novo FA (**DNFA**), 0.08 to 2.20 g/100 g milk for mixed FA (**MFA**), and 0.06 to 1.90 g/100 g milk for preformed FA (**PFFA**). Mean FA chain length and FA unsaturation were measured as previously described by Wojciechowski and Barbano (2016). Milk component yield was calculated as the sum-product of daily milk yields at each milking throughout a given day and the analyzed component values of the same day. The ECM was calculated according to equation 2 from Tyrrell and Reid (1965).

Blood. Three blood samples were drawn from each cow twice per day from d 15 to 18 of each period. Cows were sampled at 0530 h before feeding and first milking session (a.m.), and 8 h after feeding and after second milking (p.m.) from the coccygeal vessels into sodium and lithium heparinized Vacutainers and Vacutainer serum tubes (Becton Dickinson, Rutherfurd, NJ). Plasma and serum samples were separated by centrifugation at 3,100 × g for 20 min at 4°C. Plasma samples were used to analyze plasma urea N (**PUN**), glucose, insulin, and AA, and serum samples were used to analyze nonesterified fatty acids (**NEFA**). The PUN was analyzed using an enzymatic colorimetric assay based on a purchased commercial kit (no. 640, Sigma-Aldrich) as described by Chaney and Marbach (1962). Determination of glucose and NEFA concentration was conducted using enzymatic methods and commercially available kits (Autokit Glucose No. 997-03001; HR series NEFA-HR Nos. 999-34691, 995-34791, 991-34891, and 993-35191, Fujifilm, Lexington, MA). Measurement of insulin concentrations was performed using RIA (no. PI-12K Porcine Insulin RIA Kit, EMD Millipore Corp.) on an LKB-Wallac CliniGamma Counter (Beckman Coulter) as previously described by Krumm et al. (2019). Plasma AA analysis was performed as described by Ortega and Van Amburgh (2023) using Z-HILIC and triple-quadrupole LCMS. Briefly, plasma was deproteinized using equal volumes of Seraprep (Pickering Laboratories), placed on ice for 16 h at 4°C, and centrifuged at 15,700 × g for 10 min at 4°C. Supernatant was diluted 1:10 using 0.1% formic acid 90% acetonitrile (initial mobile phase) and filtered through a 2-µm syringe filter.

Power Calculation and Statistical Analysis

A prospective power analysis to determine the sample size was conducted before data collection. For the primary outcome of interest, ECM, we expected means of ~40 kg/d for the 90AA diet and 43 kg/d for the 100AA diet (Δ = 3 kg/d), as shown previously by LaPierre (2021). Assuming a pooled SD of 2.5 kg/d, with 80% power and an α of 0.05 using an F-test, the calculation indicated a minimum of 11 cows per treatment. To ensure adequate power, we rounded up to 12 cows per treatment.

Due to an illness unrelated to the experiment, data for 1 cow of the second block during period 2 were discarded. The statistical analysis was performed using R (version 4.2.0; R Core Team, 2022). All data collected over the 4-d sampling of each period was averaged for each cow and analyzed with a mixed-effects model using the functions "lme" and "anova" of the "nlme" package (Pinheiro et al., 2025). The model for production data was as follows:

$$Y_{ijkl} = \mu + Cov + Trt_i + Pd_j + Trt_i \times Pd_j + Cow(Block)_{k(l)} + e_{ijkl},$$

where Y_{ijkl} = dependent variable, μ = overall mean of Y, Cov = fixed effect of the covariate mean of Y, Trt_i =

fixed effect of the ith treatment (i = 1–5), Pd_j = fixed effect of the jth period (j = 1–5), $Trt_i \times Pd_j$ = fixed effect of the interaction between the ith treatment and the jth period, $Cow(Block)_{k(l)}$ = random effect of the kth cow nested within the lth block (k = 1–12; l = 1–2). The model for blood metabolite and hormone data was the previously described model with the addition of $Time\ of\ Day\ (ToD)_m$ = fixed effect of mth blood collection timing (m = 1–2). Treatment and time of day (**ToD**) interaction were tested for blood data, and no significant differences were observed. To account for serial correlation of repeated periods within cow, a first-order autoregressive covariance structure [**AR(1)**] was specified. For production outcomes, AR(1) was fit across periods within cow. For blood outcomes, a split AR(1) was specified by ToD (a.m. vs. p.m.), fitting separate within-cow AR(1) processes across periods for each ToD to allow the temporal correlation pattern to differ by ToD while keeping a common residual variance. Statistical significance was declared when $P \leq 0.05$ and trend toward significance as $0.05 < P \leq 0.10$. Post hoc pairwise comparison testing adjusted using Tukey's method was conducted only when the treatment effect was significant or trending toward significance, using the function "emmeans" of the "emmeans" package (Lenth et al., 2022). The study results are the least squares treatment means (LSM) and SEM. Values from chemical analyses, diet composition, and CNCPS outputs are arithmetic means and SD.

RESULTS

Diets: Composition, Chemistry, and CNCPS Predictions

The nutrient composition of the forage and concentrate diet ingredients are shown in Tables 3 and 4, respectively. The average nutrient composition and digestibility of the Cov Diet and Exp Diets are in Table 5. The AA concentrations for these ingredients have been presented in another publication (Ortega et al., 2024). Outputs from CNCPS (v.6.5.5) using the nutrient composition of these ingredients and measured animal inputs are in Table 6. As mentioned, 2 Exp Diets are presented due to a forage management constraint beginning in period 2 that can affect the supply of both feed and microbial AA, as digestibility is the primary factor for microbial AA supply. The Exp Diet 1 averaged 32.6% aNDFom (% DM) with 29.2% being uNDFom (% aNDFom), whereas Exp

Table 3. Average (±SD) nutrient composition of forage ingredients[1]

Item[2]	Corn silage (CS)	Brown midrib CS (BMR)	Grass silage 1	Grass silage 2
DM, % as-fed	39.9 ± 3.68	31.6 ± 0.82	36.7 ± 5.10	39.0 ± 1.81
CP, % DM	8.95 ± 0.22	8.47 ± 0.20	20.2 ± 1.56	18.1 ± 0.16
NDICP, % CP	8.75 ± 0.92	10.4 ± 0.68	17.0 ± 6.72	18.1 ± 0.64
ADICP, % CP	7.65 ± 0.50	8.52 ± 0.64	9.00 ± 2.55	8.33 ± 0.14
Soluble protein, % CP	66.9 ± 6.2	62.8 ± 1.27	54.2 ± 10.19	51.9 ± 1.81
RUP, % CP	16.6 ± 3.1	18.6 ± 0.63	22.9 ± 5.07	24.1 ± 0.91
Sugar, % DM	0.40 ± 0.15	1.69 ± 0.40	2.55 ± 0.50	4.43 ± 0.19
Starch, % DM	39.4 ± 0.99	28.8 ± 1.35	1.30 ± 0.29	1.38 ± 0.12
Starch digestion 7 h, % starch	79.7 ± 0.64	61.4 ± 3.37	ND	ND
aNDFom,[3] % DM	33.7 ± 1.70	34.0 ± 1.16	42.8 ± 5.73	35.3 ± 1.84
uNDFom 30 h, % DM	15.7 ± 1.14	11.0 ± 1.10	19.3 ± 3.75	9.72 ± 0.78
uNDFom 120 h, % DM	12.6 ± 1.84	7.72 ± 1.23	15.9 ± 1.63	6.39 ± 0.47
uNDFom 240 h, % DM	11.5 ± 1.49	6.57 ± 1.19	14.7 ± 1.42	5.39 ± 0.47
ADF, % DM	21.0 ± 0.64	24.6 ± 0.73	33.5 ± 3.47	32.9 ± 0.82
Lignin, % DM	2.77 ± 0.05	2.20 ± 0.10	6.4 ± 1.05	4.92 ± 0.27
Ether extract, % DM	2.86 ± 0.15	3.1 ± 0.22	3.85 ± 0.15	4.10 ± 0.11
Total fatty acids (TFA), % DM	2.56 ± 0.02	1.88 ± 0.19	1.97 ± 0.15	1.31 ± 0.03
C16:0, % TFA	17.6 ± 0.10	15.9 ± 1.13	19.6 ± 1.13	11.0 ± 0.36
C18:0, % TFA	2.74 ± 0.03	2.57 ± 0.32	3.23 ± 0.57	2.3 ± 0.02
C18:1, % TFA	22.9 ± 0.15	20.8 ± 1.77	3.33 ± 0.61	3.42 ± 0.04
C18:2, % TFA	53.7 ± 1.13	49.4 ± 1.46	19.6 ± 0.41	19.7 ± 0.42
C18:3, % TFA	1.76 ± 0.82	7.89 ± 0.69	48.4 ± 6.79	46.8 ± 1.34
Ca, % DM	0.20 ± 0.02	0.21 ± 0.02	1.07 ± 0.05	0.83 ± 0.03
Mg, % DM	0.15 ± 0.01	0.16 ± 0.02	0.29 ± 0.03	0.31 ± 0.02
P, % DM	0.26 ± 0.03	0.28 ± 0.01	0.42 ± 0.05	0.43 ± 0.02

[1]Analyses performed by Cumberland Valley Analytical Services (Waynesboro, PA), unless otherwise stated. ND = not determined.

[2]NDICP = neutral detergent insoluble crude protein; ADICP = acid detergent insoluble crude protein; uNDFom = sodium sulfite and alpha amylase–treated NDF corrected for organic matter; uNDFom 30 h, 120 h, 240 h = sodium sulfite and alpha amylase–treated NDF after 30, 120, and 240 h in-vitro rumen fermentation corrected for organic matter, termed as unavailable NDF (uNDF).

[3]aNDFom analysis was performed at Cornell University.

Table 4. Average (±SD) nutrient composition of concentrate ingredients[1]

Item	Soybean meal	Citrus pulp	Corn grain	Wheat middlings	Blood meal	AminoPlus[2]
DM, % as-fed	87.50 ± 0.70	90.88 ± 1.59	86.18 ± 0.45	86.53 ± 1.33	88.13 ± 2.93	87.30 ± 0.35
CP, % DM	52.70 ± 1.10	6.88 ± 0.41	8.68 ± 0.92	20.55 ± 0.80	99.18 ± 3.26	50.14 ± 1.06
Soluble protein, % CP	26.10 ± 2.81	39.63 ± 2.53	20.55 ± 5.43	42.58 ± 4.41	14.48 ± 2.84	18.90 ± 3.25
NDICP, % CP	1.93 ± 1.42	32.5 ± 3.71	5.53 ± 0.48	19.65 ± 3.45	4.28 ± 3.92	5.80 ± 3.21
ADICP, % CP	1.05 ± 0.52	14.15 ± 3.44	2.90 ± 0.74	3.68 ± 0.72	1.35 ± 0.95	1.97 ± 0.51
uNitrogen,[3] % total N	4.11 ± 0.40	41.20 ± 2.30	15.03 ± 3.17	5.87 ± 0.38	35.41 ± 1.83	7.00 ± 0.93
Sugar, % DM	12.93 ± 2.39	29.97 ± 1.68	2.75 ± 0.74	6.35 ± 4.00	2.40 ± 2.21	8.70 ± 7.72
Starch, % DM	0.73 ± 0.25	0.55 ± 0.21	70.78 ± 2.08	14.75 ± 2.47	0.18 ± 0.15	1.00 ± 0.20
aNDFom, % DM	8.75 ± 0.52	22.83 ± 0.46	10.60 ± 0.81	45.60 ± 3.28	5.25 ± 4.83	14.64 ± 0.86
12-h NDFd,[4] % NDF	50.67 ± 10.72	49.74 ± 1.06	41.44 ± 7.87	42.24 ± 0.67	ND	17.50 ± 0.71
72-h NDFd, % NDF	77.70 ± 2.34	73.94 ± 1.59	86.40 ± 1.42	65.50 ± 0.10	ND	85.95 ± 0.36
120-h NDFd, % NDF	81.10 ± 2.71	76.54 ± 0.77	87.04 ± 2.22	69.40 ± 0.86	ND	87.5 ± 0.15
ADF, % DM	5.95 ± 0.85	14.40 ± 2.09	3.83 ± 0.35	14.03 ± 1.03	1.75 ± 1.31	7.47 ± 1.11
Lignin, % DM	0.77 ± 0.26	2.59 ± 0.13	1.65 ± 0.20	4.52 ± 0.35	1.06 ± 0.59	0.78 ± 0.09
Ether extract, % DM	1.95 ± 0.52	2.65 ± 0.63	4.25 ± 0.34	4.79 ± 0.24	1.26 ± 1.09	1.73 ± 0.20
Ca, % DM	0.46 ± 0.15	3.11 ± 0.24	0.03 ± 0.03	0.12 ± 0.01	0.07 ± 0.05	0.39 ± 0.08
Mg, % DM	0.33 ± 0.03	0.17 ± 0.01	0.11 ± 0.01	0.62 ± 0.04	0.02 ± 0.01	0.33 ± 0.02
P, % DM	0.76 ± 0.06	0.12 ± 0.01	0.29 ± 0.03	1.25 ± 0.11	0.14 ± 0.05	0.73 ± 0.07
K, % DM	2.53 ± 0.13	0.95 ± 0.04	0.38 ± 0.03	1.32 ± 0.12	0.22 ± 0.07	2.49 ± 0.16

[1]Analyses performed by Cumberland Valley Analytical Services (Waynesboro, PA), unless otherwise stated. ND = not determined.

[2]Ag Processing Inc., Omaha, NE.

[3]Intestinally unavailable nitrogen (uNitrogen) was analyzed at Cornell University.

[4]NDFd = digestible NDF.

Diet 2 averaged 33.6% aNDFom (% DM) and 22.1% uNDFom (% aNDFom; Table 5), the potentially digestible aNDFom pool increased by 7.1 units, which will support greater microbial protein yield, thus changing the AA supply relative to the initial treatment. Similarly, the CS used in Exp Diet 1 averaged 11.5% uNDFom at 240 h (% DM) whereas the BMR used in Exp Diet 2 averaged 6.57% (Table 3). Consistently, the apparent total-tract aNDFom digestibility (% aNDFom) of Exp Diet 1 was 41.0%, and for Exp Diet 2 it was 49.5%.

To achieve the 10% difference in MP supply in the Exp Diets (Table 1), lower amounts of higher-protein ingredients and RUP, such as heat-treated SBM, bloodmeal, rumen-protected methionine and lysine, and haylage, were included in those diets. To offset the decrease in these ingredients, higher amounts of urea, wheat middlings, SBM, corn meal, and citrus pulp were added. Three separate batches of the premix were used throughout the experiment, and variability in nutrient composition was low (Table 4). The Cov Diet had a higher RUP (% CP) as predicted by the CNCPS, 38.6%, compared with Exp Diets 1 and 2, 35.3% and 35.7%, respectively (Table 6). The opposite was observed for predictions of RDP (% CP) where the Cov Diet had lower levels than the Exp Diets. The MP predictions for the Cov Diet, Exp Diet 1, and Exp Diet 2 were 3,032 g/d, 2,797 g/d, and 2,842 g/d, respectively. Individual EAA and total NEAA followed the same trend (Table 6). The MP allowable milk decreased more than 10%, from 41.8 kg/d in the Cov Diet to 36.0 kg/d in Exp Diet 1 and 37.0 kg/d in Exp Diet 2. Estimated microbial MP was similar between Exp Diets 1 and 2 at 59.3% and 59.5% MP, respectively. Diets were formulated to be isocaloric, which was achieved between Cov Diet and Exp Diet 1, having the same ME of 2.69 Mcal/kg, but ME increased in Exp Diet 2 to 2.75 Mcal/kg (Table 6).

Animal Performance and Production

Milk production and concentrations, animal measurements, DMI, and feed efficiency are shown in Table 7. Results were similar among treatments (**Trt**) for BW (Trt $P = 0.18$), BCS (Trt $P = 0.65$), DMI (Trt $P = 0.38$), Milk yield (Trt $P = 0.98$), ECM (Trt $P = 0.38$), and milk true protein (Trt $P = 0.38$). The concentration of MUN increased when higher amounts of AA were infused (Trt $P < 0.01$). Milk fat concentration (%; Trt $P = 0.10$) and milk fat yield (kg/d; Trt $P = 0.10$) trended to decrease in the 110NEAA treatment.

Given that both milk fat concentration and yield were trending toward significance, we further evaluated milk FA metrics estimated from FTIR spectra (Table 8). Consistent with the previous results, total FA (**TFA**) concentration (g/100 g milk; Trt $P = 0.02$) and yield (g/d; Trt $P = 0.04$) were lowest for the 110NEAA treatment at 4.14 g/100 g milk and 1,736 g/d and highest for the 100EAA treatment (4.45 g/100 g milk and 1,857 g/d) and the 100AA treatment (4.45 g/100 g milk and 1,842 g/d). The lower milk fat response in 110NEAA reflected lower TFA concentration (g/100 g milk) and lower TFA yield (g/d), driven primarily by PFFA. All metrics related to PFFA (i.e., g/100 g milk, g/d, and g/100 g FA) were sig-

Table 5. Average (±SD) nutrient composition and digestibility of the covariate (Cov Diet) and experimental diets (Exp Diet)[1,2]

Item	Cov Diet (Covariate period)	Exp Diet 1 (Period 1)	Exp Diet 2 (Periods 2–5)
DM, % as-fed	49.92 ± 1.4	50.0 ± 1.2	45.8 ± 2.0
CP, % DM	16.4 ± 1.5	15.8 ± 0.6	14.9 ± 0.2
NDICP, % CP	18.0 ± 2.0	17.2 ± 1.7	18.8 ± 0.5
ADICP, % CP	6.68 ± 0.3	7.22 ± 0.4	7.07 ± 0.1
Soluble protein, % CP	43.5 ± 3.9	47.8 ± 3.4	47.8 ± 1.5
RUP, % CP	28.2 ± 2.0	26.1 ± 1.7	26.1 ± 0.8
Sugar, % DM	7.00 ± 1.8	7.04 ± 1.7	8.34 ± 0.2
Starch, % DM	26.7 ± 0.8	25.8 ± 2.0	22.2 ± 1.2
aNDFom,[3] % DM	28.8 ± 1.3	32.6 ± 1.3	33.6 ± 1.4
uNDFom 240 h,[3] % aNDFom	31.8 ± 2.0	29.2 ± 1.1	22.1 ± 1.3
aTTD,[3,4] % aNDFom	53.2 ± 4.9	41.0 ± 7.9	49.5 ± 7.2
ADF, % DM	18.1 ± 1.4	18.9 ± 1.5	20.3 ± 0.8
Lignin, % DM	2.84 ± 0.3	2.86 ± 0.3	2.45 ± 0.1
Ether extract, % DM	4.58 ± 0.4	4.56 ± 0.2	4.60 ± 0.1
C18:0, % total FA	7.23 ± 0.4	7.36 ± 0.6	7.80 ± 0.1
C18:1, % total FA	13.8 ± 1.7	16.3 ± 1.7	17.0 ± 0.6
C18:2, % total FA	17.3 ± 5.2	18.0 ± 2.8	18.8 ± 1.4
C18:3, % total FA	13.4 ± 2.7	14.4 ± 2.5	17.8 ± 1.6
NEL, Mcal/kg	1.70 ± 0.01	1.70 ± 0.01	1.70 ± 0.01
Ca, % DM	0.77 ± 0.05	0.69 ± 0.1	0.62 ± 0.03
Mg, % DM	0.30 ± 0.04	0.27 ± 0.01	0.27 ± 0.01
P, % DM	0.37 ± 0.03	0.37 ± 0.02	0.36 ± 0.01
K, % DM	1.70 ± 0.2	1.80 ± 0.1	1.78 ± 0.07
Ash, % DM	7.83 ± 0.4	7.34 ± 0.3	6.98 ± 0.2

[1]Analyses performed by Cumberland Valley Analytical Services (Waynesboro, PA), unless otherwise stated.

[2]All cows were fed the same diet during the corresponding period, regardless of treatment.

[3]aNDFom, uNDFom 240 h, and aTTD analysis were conducted at Cornell University.

[4]aTTD = apparent total-tract digestibility.

Table 6. Metabolizable energy and protein allowable milk and metabolizable protein content and supply, rumen characteristics and AA supplies as predicted by the Cornell Net Carbohydrate and Protein System (prediction ± SD) using AMTS Cattle Professional (AMTS LLC; v. 4.16.6) for the covariate (Cov Diet) and experimental diets (Exp Diet)

Item	Cov Diet (Covariate period)	Exp Diet 1 (Period 1)	Exp Diet 2 (Periods 2–5)
DMI, kg/d	26.6	27.5	27.4 ± 0.7
ME, Mcal/kg	2.69	2.69	2.75 ± 0.01
ME, % required	92.8	96.3	97.2 ± 2.4
ME allowable milk, kg/d	40.6	42.8	43.3 ± 1.5
MP, g/d	3,032	2,797	2,843 ± 43
MP, % required	95.0	86.1	87.7 ± 1.4
MP allowable milk, kg/d	41.8	36.0	37.0 ± 0.9
Microbial MP, % MP	55.4	59.3	59.5 ± 0.8
RUP, % CP	38.6	35.3	35.7 ± 0.4
RDP, % CP	61.4	64.4	64.3 ± 0.4
Arg, g/d	187.7	173.6	176.3 ± 2.7
His, g/d	85.8	77.2	78.2 ± 1.3
Ile, g/d	144.2	136.1	138.5 ± 1.9
Leu, g/d	241.8	221.0	224.6 ± 3.6
Lys, g/d	222.0	202.3	205.0 ± 3.0
Met, g/d	83.5	74.0	74.9 ± 1.2
Phe, g/d	153.8	140.6	142.8 ± 2.2
Thr, g/d	144.9	135.0	137.2 ± 2.0
Trp, g/d	43.6	40.2	40.8 ± 0.6
Val, g/d	174.9	160.8	163.2 ± 2.4
Total EAA MP, g/d	1,482	1,361	1,381 ± 21
Total NEAA MP, g/d[1]	1,550	1,437	1,461 ± 23

[1]Total NEAA MP calculated by the difference between total MP and total EAA MP.

nificantly lower ($P < 0.05$) for the 110NEAA treatment compared with the 100EAA treatment (Table 8). This pattern was also evident for individual PFFA. In 110NEAA, C18:0 concentration (g/100 g milk) and yield (g/d) were lower (Trt $P < 0.01$ for both), and C18:1 *cis*-9 concentration and yield were lower (Trt $P = 0.01$ for both) relative to treatments with higher PFFA. Comparably, milk concentration (g/100 g milk) was lower for MFA (Trt $P = 0.04$) and trending lower for DNFA (Trt $P = 0.10$) in the 110NEAA treatment. In contrast, the FA proportion (g/100 g FA) of MFA (Trt $P = 0.01$) was higher with the 110NEAA treatment (42.3 g/100 g FA) compared with the 100EAA treatment (41.4 g/100 g FA).

Blood AA, Metabolites, Energy Nutrients, and Insulin

The plasma concentrations analyzed for EAA and NEAA are shown in Tables 9 and 10, respectively. The concentrations of total EAA differed among treatments, with higher concentrations observed when EAA was infused (Trt $P < 0.01$). Of the individual EAA, Leu, Lys, Met, Phe, and Val were significantly different (Trt $P \leq 0.05$), and Thr and Trp were trending toward significance (Trt $0.05 < P \leq 0.1$) among treatments, whereas Arg, His, and Ile were not (Trt $P > 0.10$). Of interest, the 100EAA treatment had higher Lys concentrations (44.7 μ*M*) than the 100AA (35.6 μ*M*) and 110NEAA treatments (36.8 μ*M*), even though the same quantity of Lys was infused. Lower concentrations of Arg, His, Lys, Thr, and Trp were observed when blood was collected in the p.m. compared with a.m. (ToD $P \leq 0.05$), whereas the remaining EAA were unaffected by time (ToD $P > 0.10$). Total NEAA, Asn, Gln, Glu, Pro, and Ser were affected by treatment (Trt P 0.05), but Ala, free Cystine (Cys2), Gly, and Tyr were not (Trt $P > 0.10$). Despite the 110NEAA treatment being infused with twice the amount of NEAA than the 100NEAA and 100AA treatments, the plasma concentrations of most NEAA were similar between these treatments except for Pro. The concentration of Pro was higher for the 110NEAA (84.2 μ*M*) compared with the 100NEAA (77.5 μ*M*) and 100AA treatments (75.6 μ*M*). Other than Cys2, Glu, and Tyr (ToD $P > 0.10$), all plasma NEAA and total NEAA were affected by collection ToD ($P < 0.10$). Compared with the EAA, most of the plasma NEAA and total NEAA were lower for p.m. collection.

The AA metabolites, energy nutrients, and insulin blood concentrations are shown in Table 11. From the metabolites related to muscle protein, 1-methylhistidine (**1MH**) and 3-methylhisitidine (**3MH**) were significantly different among treatments (Trt $P < 0.01$), whereas

Table 7. Effects of abomasal infusion of AA on lactation performance, BW, BCS, and DMI (LSM)

	Treatment[1]						*P*-value[2]		
Item	90AA	100EAA	100NEAA	100AA	110NEAA	SEM	Trt	Pd	Trt × Pd
Milk production, kg/d									
ECM[3]	48.4	48.3	47.8	48.4	47.2	1.1	0.38	0.04	0.10
Milk yield	42.2	42.2	41.9	42.0	42.1	1.0	0.98	<0.01	0.56
Fat yield	1.91	1.91	1.85	1.90	1.80	0.07	0.10	0.22	0.25
True protein yield	1.31	1.31	1.31	1.32	1.31	0.03	0.92	0.58	0.37
Lactose yield	1.93	1.93	1.92	1.93	1.94	0.05	0.99	<0.01	0.16
Solids yield	5.67	5.69	5.63	5.68	5.57	0.14	0.64	0.01	0.04
SNF yield	3.71	3.70	3.69	3.72	3.71	0.08	0.98	<0.01	0.36
Milk concentration, %									
Fat	4.54	4.59	4.49	4.58	4.35	0.13	0.10	<0.01	0.27
True protein	3.11	3.12	3.13	3.18	3.17	0.05	0.38	<0.01	0.47
Lactose	4.60	4.58	4.61	4.60	4.60	0.02	0.08	0.04	0.30
Solids	13.47	13.57	13.53	13.64	13.35	0.23	0.29	<0.01	0.24
SNF	8.81	8.80	8.84	8.88	8.88	0.07	0.14	<0.01	0.27
MUN, mg/dL	8.28[a]	8.92[a]	8.88[a]	10.03[b]	10.56[b]	0.32	<0.01	<0.01	0.64
Body measurements, intake, and efficiency, kg/d									
BW	679	706	706	717	723	13	0.18	0.93	0.30
BCS, 1–5 scale	2.79	2.83	2.81	2.88	2.85	0.04	0.65	0.86	0.13
DMI	27.6	27.4	27.6	27.1	27.2	0.7	0.38	0.02	0.52
ECM feed efficiency	1.77	1.79	1.75	1.81	1.74	0.04	0.07	<0.01	0.28

[a,b]Means within a row differ with different superscripts ($P \leq 0.05$).

[1]90AA = water, no AA infused; 100EAA = EAA infusion at 184.83 g/d; 100NEAA = NEAA infusion at 158.22 g/d; 100AA = EAA and NEAA infusion at 344.14 g/d; 110NEAA = EAA and 2× NEAA infusion at 501.91 g/d.

[2]Trt = treatment; Pd = period.

[3]Calculated according to Tyrrell and Reid (1965) equation 2: ECM = (MY × 0.327) + (Fat Yield × 12.97) + (Protein Yield × 7.65).

Table 8. Effects of abomasal infusion of AA on total, de novo, mixed, and preformed fatty acid (FA) production (LSM)

	Treatment[1]						*P*-value[2]		
Item	90AA	100EAA	100NEAA	100AA	110NEAA	SEM	Trt	Pd	Trt × Pd
Total fatty acids									
g/100 g milk	4.35[ab]	4.45[a]	4.37[ab]	4.45[a]	4.14[b]	0.16	0.02	<0.01	0.29
g/d	1,826[ab]	1,857[a]	1,798[ab]	1,842[a]	1,736[b]	73	0.04	0.01	0.07
De novo fatty acids[3]									
g/100 g milk	1.20	1.23	1.21	1.24	1.16	0.05	0.10	<0.01	0.13
g/d	504	510	498	511	488	25	0.35	<0.01	0.07
g/100 g FA	27.2	27.2	27.4	27.5	27.7	0.2	0.27	<0.01	0.81
Mixed fatty acids[3]									
g/100 g milk	1.84[ab]	1.87[a]	1.85[ab]	1.88[a]	1.77[b]	0.07	0.04	<0.01	0.21
g/d	772	775	758	777	743	43	0.18	<0.01	0.05
g/100 g FA	41.7[ab]	41.4[a]	41.7[ab]	42.0[ab]	42.3[b]	0.3	0.01	<0.01	0.02
Preformed fatty acids[3]									
g/100 g milk	1.37[ab]	1.40[a]	1.36[ab]	1.36[ab]	1.26[b]	0.07	0.02	0.80	0.47
g/d	574[a]	591[a]	564[ab]	566[ab]	527[b]	27	0.01	0.01	0.19
g/100 g FA	31.2[a]	31.3[a]	31.0[ab]	30.6[ab]	30.0[b]	0.4	0.01	<0.01	0.15
Fatty acids									
C16:0, g/100 g milk	1.73[ab]	1.76[ab]	1.74[ab]	1.77[a]	1.66[b]	0.07	0.05	<0.01	0.24
C16:0, g/d	727	730	713	731	698	42	0.18	<0.01	0.06
C18:0, g/100 g milk	0.42[a]	0.42[a]	0.41[ab]	0.41[ab]	0.37[b]	0.03	<0.01	0.72	0.44
C18:0, g/d	174.6[a]	178.5[a]	170.3[ab]	168.0[ab]	156.1[b]	11.8	<0.01	0.20	0.10
C18:1 *cis*-9, g/100 g milk	0.73[ab]	0.75[a]	0.72[ab]	0.74[ab]	0.67[b]	0.02	0.01	0.14	0.30
C18:1 *cis*-9, g/d	306.4[ab]	315.1[a]	299.3[ab]	308.3[ab]	282.9[b]	8.3	0.01	<0.01	0.13

[a,b]Means within a row differ with different superscripts ($P \leq 0.05$).

[1]90AA = water, no AA infused; 100EAA = EAA infusion at 184.83 g/d; 100NEAA = NEAA infusion at 158.22 g/d; 100AA = EAA and NEAA infusion at 344.14 g/d; 110NEAA = EAA and 2× NEAA infusion at 501.91 g/d.

[2]Trt = treatment; Pd = period.

[3]De novo = C4:0 to C14:0; mixed = C16:0, C16:1, C17:0; preformed = C18:0 and longer (Barbano et al., 2014).

Table 9. Effects of abomasal infusion of AA on EAA concentration (LSM) before feeding and first milking (a.m.) and 8 h after feeding and after second milking (p.m.)

		Treatment[1]						*P*-value[2]			
AA, μ*M*	Time	90AA	100EAA	100NEAA	100AA	110NEAA	SEM	Trt	Pd	Trt × Pd	ToD
Arg	a.m.	44.1	45.3	43.4	43.8	46.5	2.6	0.58	0.05	0.22	<0.01
	p.m.	39.0	40.2	38.4	38.7	41.4					
His	a.m.	29.3	30.6	28.5	30.2	27.6	1.9	0.18	0.68	0.08	0.03
	p.m.	27.3	28.5	26.4	28.2	25.5					
Ile	a.m.	81.4	83.8	80.5	81.6	83.5	3.8	0.63	0.03	0.64	0.66
	p.m.	82.6	85.1	81.8	82.9	84.8					
Leu	a.m.	120.1[a]	138.4[b]	118.6[a]	133.8[ab]	139.8[b]	5.8	<0.01	<0.01	0.37	0.49
	p.m.	123.9[a]	142.3[b]	122.5[a]	137.7[ab]	143.7[b]					
Lys	a.m.	40.4[ab]	46.2[a]	39.7[ab]	37.1[b]	38.3[b]	2.5	0.01	0.87	0.05	0.03
	p.m.	37.4[ab]	43.1[a]	36.7[ab]	34.1[b]	35.2[b]					
Met	a.m.	25.9[a]	32.1[b]	25.8[a]	32.0[b]	30.0[b]	1.4	<0.01	<0.01	0.03	0.51
	p.m.	26.5[a]	32.6[b]	26.3[a]	32.5[b]	30.6[b]					
Phe	a.m.	39.0[a]	45.0[b]	40.1[a]	46.5[bc]	51.0[c]	1.9	<0.01	0.10	0.03	0.78
	p.m.	39.5[a]	45.4[b]	40.5[a]	46.9[bc]	51.4[c]					
Thr	a.m.	87.4[a]	87.4[a]	77.7[b]	84.0[ab]	83.7[ab]	4.4	0.07	<0.01	0.02	<0.01
	p.m.	75.9[a]	75.9[a]	66.2[b]	72.4[ab]	72.2[ab]					
Trp	a.m.	32.6	33.8	31.3	32.1	32.3	1.4	0.08	0.03	0.05	0.01
	p.m.	29.7	30.9	28.4	29.2	29.4					
Val	a.m.	197.6[a]	229.1[b]	196.8[a]	219.6[b]	229.7[b]	7.8	<0.01	<0.01	0.33	0.94
	p.m.	197.2[a]	228.7[b]	196.5[a]	219.3[b]	229.3[b]					
EAA	a.m.	698.8[ac]	773.4[b]	683.2[c]	739.4[abc]	762.6[ab]	24.4	<0.01	0.01	0.34	0.40
	p.m.	679.6[ac]	754.2[b]	664.0[c]	720.2[abc]	743.3[ab]					

[a–c]Means within a row differ with different superscripts ($P \leq 0.05$).

[1]90AA = water, no AA infused; 100EAA = EAA infusion at 184.83 g/d; 100NEAA = NEAA infusion at 158.22 g/d; 100AA = EAA and NEAA infusion at 344.14 g/d; 110NEAA = EAA and 2× NEAA infusion at 501.91 g/d.

[2]Trt = treatment; Pd = period; ToD = time of day: before feeding and first milking session (a.m.) and 8 h after feeding and after second milking (p.m.).

Table 10. Effects of abomasal infusion of AA on NEAA concentration (LSM) before feeding and first milking (a.m.) and 8 h after feeding and after second milking (p.m.)

		Treatment[1]						*P*-value[2]			
AA, μ*M*	Time	90AA	100EAA	100NEAA	100AA	110NEAA	SEM	Trt	Pd	Trt × Pd	ToD
Ala	a.m.	244.8	243.6	253.2	243.6	250.3	11.6	0.50	0.08	0.18	<0.01
	p.m.	207.4	206.3	215.8	206.2	212.9					
Asn	a.m.	48.0[a]	47.7[a]	53.7[ab]	53.6[ab]	60.7[b]	2.4	<0.01	0.27	0.66	<0.01
	p.m.	38.9[a]	38.6[a]	44.6[ab]	44.5[ab]	51.7[b]					
Cys2[3]	a.m.	11.7	11.1	11.4	11.0	11.9	1.0	0.95	0.54	0.26	0.49
	p.m.	11.2	10.6	10.8	10.4	11.3					
Gln	a.m.	179.3[ab]	167.9[a]	194.8[b]	188.7[b]	193.1[b]	15.2	<0.01	0.03	0.92	<0.01
	p.m.	146.9[ab]	135.6[a]	162.4[b]	156.4[b]	160.7[b]					
Glu	a.m.	34.5[a]	32.3[ab]	34.8[a]	30.6[b]	33.0[ab]	1.8	0.04	0.47	0.57	0.48
	p.m.	35.7[a]	33.5[ab]	36.0[a]	31.9[b]	34.2[ab]					
Gly	a.m.	338.3	303.9	348.9	322.3	336.0	34.9	0.21	0.84	0.44	<0.01
	p.m.	265.2	230.8	275.8	249.1	262.8					
Pro	a.m.	71.4[a]	71.0[a]	83.7[b]	81.8[b]	90.4[b]	4.3	<0.01	0.59	0.49	<0.01
	p.m.	59.0[a]	58.6[a]	71.3[b]	69.4[b]	78.0[b]					
Ser	a.m.	70.1[ab]	68.3[a]	77.2[ab]	73.0[ab]	77.6[b]	3.5	<0.01	0.82	0.33	<0.01
	p.m.	59.1[ab]	57.4[a]	66.3[ab]	62.0[ab]	66.6[b]					
Tyr	a.m.	35.8	36.8	36.3	38.5	38.2	1.8	0.41	<0.01	0.07	0.47
	p.m.	34.9	35.8	35.4	37.6	37.3					
NEAA	a.m.	1,042.0[ab]	990.1[a]	1,103.4[b]	1,053.5[ab]	1,101.6[b]	42.3	0.01	0.28	0.51	<0.01
	p.m.	846.5[ab]	794.7[a]	908.0[b]	858.0[ab]	906.1[b]					

[a,b]Means within a row differ with different superscripts ($P \leq 0.05$).

[1]90AA = water, no AA infused; 100EAA = EAA infusion at 184.83 g/d; 100NEAA = NEAA infusion at 158.22 g/d; 100AA = EAA and NEAA infusion at 344.14 g/d; 110NEAA = EAA and 2× NEAA infusion at 501.91 g/d.

[2]Trt = treatment; Pd = period; ToD = time of day: before feeding and first milking session (a.m.) and 8 h after feeding and after second milking (p.m.).

[3]Cys2 = free cystine.

Table 11. Effects of abomasal infusion of AA on AA metabolites, energy nutrients, and insulin concentration (LSM) before feeding and first milking (a.m.), and 8 h after feeding and after second milking (p.m.)

Item[3]	Time	Treatment[1]					SEM	*P*-value[2]			
		90AA	100EAA	100NEAA	100AA	110NEAA		Trt	Pd	Trt × Pd	ToD
1MH, μ*M*	a.m.	3.69[ac]	3.27[b]	3.58[c]	3.41[abc]	3.34[bc]	0.13	<0.01	0.04	<0.01	0.18
	p.m.	3.60[ac]	3.18[b]	3.48[c]	3.32[abc]	3.24[bc]					
3MH, μ*M*	a.m.	3.38[a]	2.74[b]	3.46[a]	3.14[abc]	3.21[a]	0.15	<0.01	0.41	0.11	0.44
	p.m.	3.31[a]	2.67[b]	3.38[a]	3.07[abc]	3.14[a]					
Carnosine, μ*M*	a.m.	9.48	8.50	9.24	9.13	9.14	0.52	0.43	<0.01	0.23	<0.01
	p.m.	8.09	7.12	7.86	7.75	7.76					
Cit, μ*M*	a.m.	52.3[ab]	50.3[a]	52.0[ab]	54.8[ab]	57.2[b]	2.6	0.01	0.01	0.67	<0.01
	p.m.	44.8[ab]	42.8[a]	44.5[ab]	47.4[ab]	49.8[b]					
Creatinine, μ*M*	a.m.	54.8	53.0	54.0	53.4	52.6	3.3	0.68	0.02	0.02	0.79
	p.m.	55.0	53.2	54.2	53.6	52.8					
Glucose, mg/dL	a.m.	67.4	69.5	66.8	65.2	67.2	2.4	0.66	0.01	0.33	0.16
	p.m.	64.5	66.6	63.9	62.3	64.3					
Insulin, ng/mL	a.m.	0.56	0.57	0.56	0.59	0.73	0.08	0.08	0.31	0.53	<0.01
	p.m.	1.01	1.01	1.00	1.03	1.17					
NEFA, μEq/L	a.m.	107.3	115.1	121.0	109.7	107.8	8.8	0.67	<0.01	0.07	0.89
	p.m.	104.5	112.3	118.2	106.9	105.0					
OH-Pro, μ*M*	a.m.	54.6	55.7	56.3	56.1	55.4	1.6	0.86	0.09	0.11	0.08
	p.m.	52.3	53.4	54.0	53.8	53.1					
Orn, μ*M*	a.m.	23.5[ab]	21.2[a]	21.7[ab]	22.1[ab]	25.7[b]	1.6	0.05	<0.01	0.11	<0.01
	p.m.	20.3[ab]	18.0[a]	18.5[ab]	18.9[ab]	22.5[b]					
PUN, mg/dL	a.m.	7.9[a]	8.8[ab]	8.5[ab]	9.5[bc]	10.5[c]	0.5	<0.01	0.01	0.63	<0.01
	p.m.	9.2[a]	10.1[ab]	9.8[ab]	10.8[bc]	11.8[c]					

[a–c]Means within a row differ with different superscripts ($P \leq 0.05$).

[1]90AA = water, no AA infused; 100EAA = EAA infusion at 184.83 g/d; 100NEAA = NEAA infusion at 158.22 g/d; 100AA = EAA and NEAA infusion at 344.14 g/d; 110NEAA = EAA and 2× NEAA infusion at 501.91 g/d.

[2]Trt = treatment; Pd = period; ToD = time of day: before feeding and first milking session (a.m.) and 8 h after feeding and after second milking (p.m.).

[3]1MH = 1-methyl histidine; 3MH = 3-methyl histidine; OH-Pro = hydroxy-proline.

carnosine and creatinine did not differ (Trt $P > 0.10$). Except for carnosine, these metabolites were not affected by blood collection ToD ($P > 0.10$). The concentration of both 1MH and 3MH were highest in the 90AA (3.65 μ*M* and 3.35 μ*M*, respectively) and 100NEAA treatments (3.53 μ*M* and 3.42 μ*M*, respectively), where no EAA were infused, and lowest in the 100EAA treatment (3.23 μ*M* and 2.71 μ*M*, respectively). Cit and Orn, AA involved in the urea cycle, were affected by treatment (Trt $P = 0.01$ and $P = 0.05$, respectively) and blood collection ToD ($P < 0.01$ for both). The highest concentrations for both Cit and Orn were measured with the 110NEAA treatment (48.3 μ*M* and 20.1 μ*M*, respectively) and were 15% lower during the p.m. blood collection than the a.m. Correspondingly, PUN (Trt $P < 0.01$) was highest in the 110NEAA treatment (11.2 mg/dL), and higher amounts were measured in the p.m. (ToD $P < 0.01$). As expected, PUN values were higher than MUN, and the same changes were observed for both, increasing when more AA were infused. The concentrations of NEFA were not different between treatments (Trt $P = 0.67$); the 110NEAA (101.7 μEq/L) and 90AA treatments (104.7 μEq/L) were numerically lower compared with the 100EAA (115.6 μEq/L), 100NEAA (117.5 μEq/L), and 100AA (110.3 μEq/L) treatments; and they were also not affected by blood collection ToD ($P = 0.89$). Finally, glucose concentrations were unrelated to treatment (Trt $P = 0.66$), but there was a trend toward higher insulin (Trt $P = 0.08$) in the 110NEAA (0.95 ng/mL) compared with the 90AA, 100EAA, and 100NEAA treatments (0.79 ng/mL for all). The ToD for blood collection only affected insulin ($P < 0.01$) but not glucose ($P = 0.16$). Insulin levels were higher during the p.m. collection than during the a.m. collection (ToD $P > 0.01$).

DISCUSSION

The objective of this study was to investigate the productive and metabolic responses to varying the supplies of EAA and NEAA to lactating dairy cattle being fed a diet that meets 100% ME and 90% MP requirements. The ME was formulated to be similar in all diets, and predictions with the observed model inputs reinforced this, although predictions of Exp Diet 2 showed a higher ME due to the changes in forages. The change in forages after the first period is noticeable from the chemical analyses as BMR had greater NDF digestibility compared to CS, thus providing more energy and bacterial growth (Miller et al., 2020). The starch content of the diet was lower than formulated in the first period, and this likely offset some of the differences in aNDFom digestibility. Given the higher digestibility of BMR, Exp Diet 2 had higher predictions of grams MP supply; thus the diet was not deficient in MP as originally formulated for the first period. This likely confounded our treatments, as with higher aNDFom digestibility and more bacterial yield, the differences among treatments were likely not as great as what was observed in the first period on Exp Diet 1. Thus, the sensitivity to the infusions was diminished, as Exp Diet 2 was not as deficient in MP as originally formulated.

Milk yield for the present study was nearly identical for all treatments. Previous studies, with cows at similar DIM, reported varying results in milk yield when both EAA and NEAA were infused. Metcalf et al. (1996) did not measure any changes in milk yield, but Doepel and Lapierre (2010) reported an increase. Studies with cows in early lactation had a significant increase in milk yield and ECM when casein or all AA were infused (Larsen et al., 2014; Galindo et al., 2015; Chandler et al., 2022). Early-lactation postpartum cows have a much higher demand for energy, which could be provided by the AA being infused. Metcalf et al. (1996) showed the highest milk protein percentage for the treatment that only had EAA infused, whereas Doepel and Lapierre (2010) reported the highest value for the treatment where all AA were infused. In our study, milk protein percent was not statistically different among treatments, likely due to the aforementioned shift in MP by diets. This was unexpected for the 110NEAA treatment, as there should be a higher availability of NEAA as well as sufficient EAA for protein synthesis. One example is plasma Pro, which increased significantly (Trt $P > 0.01$) in this study and is typically produced from Arg in the MG (Trottier et al., 1997; Lapierre et al., 2012; Wu, 2021). The NEAA themselves can also be used extensively for the synthesis of milk casein and lactose (Black et al., 1968), such as Glu and Gln, which were influenced by AA treatment in the present study. The branched-chain AA, Glu, and Pro account for more than 40% of the AA in milk protein (Lapierre et al., 2012). Similar to previous infusion studies (Metcalf et al., 1996; Doepel and Lapierre, 2010), milk fat yield and milk fat percentage trended lower. Whereas the lowest milk fat percentage in previous infusion studies was observed with infusion of all AA, in the present study, we observed lower milk fat with NEAA infusion at 110% of estimated requirements (110 NEAA treatment).

The TFA percent was highest for the 100EAA treatment and lowest for 110NEAA. Barbano et al. (2014) reported higher DNFA, MFA, and PFFA, in grams per 100 g of milk (%), with higher bulk tank milk fat percentage, but these relationships do not seem to hold true in the present study, where cows were fed an isocaloric diet and varying amounts of different AA mixes. The 110NEAA treatment had a greater proportion of MFA compared with the 100EAA treatment, whereas the inverse relationship is observed with PFFA. The MFA can come from either dietary fat or de novo synthesis, and PFFA are derived from dietary sources and endogenous FA catabolized from adipose tissue (Barbano et al., 2014; Palmquist and Harvatine, 2020). Because cows received the same diet, MFA should be coming from DNFA and PFFA from fat mobilized from adipose tissue. Consistent with these results, when postpartum cows were infused with a mix of all AA, there was an increase in both DNFA and MFA and a decrease in PFFA (Chandler et al., 2022). The higher relative proportion (g/100 g FA) of DNFA and MFA in the 110NEAA treatment could be related to higher AA availability, as all EAA and theoretical NEAA requirements were met. The excess of NEAA could have been catabolized into other substrates such as tricarboxylic acid cycle intermediates to produce more reducing equivalents (NAD(P)H, $FADH_2$) and acetyl-CoA for production of DNFA. In ruminants, optimal rumen function supports acetate production, which, along with glucose, provides C (via acetyl-CoA) for de novo synthesis of FA and subsequent elongation (Palmquist and Harvatine, 2020). Lipogenesis depends on acetyl-CoA carboxylase, a biotin-dependent enzyme in which biotin is covalently attached to a Lys residue, providing a plausible biochemical link between Lys metabolism and lipogenic capacity (Cronan, 2001; Barbano et al., 2014; Palmquist and Harvatine, 2020). In the present study, plasma Lys was lower in the 100AA and 110NEAA than in 100EAA, suggesting greater Lys utilization or disposal to other metabolic fates, including protein synthesis, under conditions of higher NEAA supply. Because diets were identical, the tendency for a higher proportion of DNFA and higher MFA observed in the 110NEAA is consistent with altered metabolic partitioning and potentially greater lipogenesis. However, our data does not allow us to directly test whether changes in Lys availability or disposal influence mammary lipogenic pathways. Additional work will be needed to evaluate whether Lys metabolism meaningfully interacts with lipogenesis under these conditions. In mice, it has been demonstrated that AA serve as a major, and usually underestimated, source of C for hepatic DNFA synthesis, accounting for ~45% of the C from dietary and synthesized AA (Liao et al., 2024). Finally, in this study, the lower PFFA values in the 110NEAA treatment may be consistent with the tendency for higher insulin concentrations in that treatment, as insulin is associated with lower lipolysis.

The insulin response was numerically highest with the 110NEAA treatment compared with the other treatments. And, although twice the amount of NEAA was being infused with 110NEAA, the NEAA levels in plasma were similar to the other treatments, except the 100EAA treatment. This implies that the additional NEAA were likely catabolized into metabolic substrates that could not be quantified in the current study. Ala plays major glucoregulatory roles in the body and can be used for gluconeogenesis (Overton et al., 1999; Wu, 2009). In fact, in sheep as much as 26% of glucose C can be derived from Ala (Black et al., 1955; Wolff and Bergman, 1972), although in dairy cattle it has been recorded at much lower levels, with higher amounts if the animal is fasted (Larsen and Kristensen, 2013). Ala did not differ among treatments in the present study, indicating that Ala is used when higher quantities are provided. This is further supported by the 110NEAA treatment having the highest PUN levels, which is a byproduct of AA catabolism (Martineau et al., 2017). Arg plays a major role in N metabolism by being a key player in the urea cycle. Arg was not found to be significantly different in this study, but its urea cycle products, Orn and Cit, were significantly higher when more AA were being infused. However, EAA concentrations differed among treatments as expected, with the 3 treatments that received EAA (100EAA, 100AA, and 110NEA) having higher values than the other 2 (90AA and 100NEAA).

Given that most studies involving AA infusions are conducted at steady state—that is, feed is supplied at constant intervals throughout the day—comparisons of AA concentrations at different times of day have not been fully investigated in infusion studies. Recently, Salfer et al. (2023) examined the daily rhythm effects of infusing casein continuously, only during the day and only at night, by sampling cows at 6 different points every 4 h, but did not measure AA. In the current study, 2 blood samples were taken to investigate that relationship. The first sample (a.m.) was taken before cows were fed and before the first milking, indicating that the cows were fasted and that minimal nutrients were diverted to milk production. The second sample (p.m.) was collected 8 h after feeding and after the second milking; thus, cows were in a postprandial state, and the MG was beginning to increase milk production due to the reduction in intramammary pressure. As expected, insulin levels were much lower during the a.m. than during the p.m., as the animals were fasted. Accordingly, the body increases glucose uptake and begins catabolizing other nutrients for energy. Salfer et al. (2023) also observed similar glucose and insulin patterns with continuous infusions, but these remained stable when the infusion was administered during the day. Toledo et al. (2021) analyzed plasma AA over a 24-h period, comparing a control diet against one with top-dressed Met, and found significant effects of time with most AA except Met, likely due to the experiment's treatment. The highest amount of plasma AA was measured at 24 h after feeding, consistent with the morning increase of AA found in this study. Many of these AA can be used as precursors for energy metabolites, which is related to the higher concentrations of AA observed in the a.m. blood samples. Additionally, because the MG has not been emptied at a.m. collection, intramammary pressure is greater, which reduces milk production and, consequently, nutrient use for this purpose (Weaver and Hernandez, 2016).

CONCLUSIONS

Abomasal infusions of NEAA beyond the estimated AA levels of predicted MP trended to reduce milk fat production, driven by a significant reduction in PFFA. The reduction in PFFA may be related to lower adipose tissue turnover because of a trend in insulin response when NEAA were infused above predicted MP requirements. Overall, diets formulated using CNCPS were designed to meet the ME and MP requirements and supply, and the observed production responses were broadly consistent with these targets across treatments. These results suggest that altering NEAA supply, independent of EAA supply, can shift milk fat partitioning in ways not captured by MP supply alone. This supports continued development of nutrition models that more explicitly track NEAA supply and utilization when predicting nutrient partitioning outcomes. The negative responses in milk FA due to additional NEAA infusions show the important metabolic interactions between FA and AA that need to be further assessed.

NOTES

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Nonstandard abbreviations used: 90AA = water-only control, meeting 90% requirement of EAA and NEAA; 100AA = EAA and NEAA mix, 344.1 g/d, meeting 100% requirement of EAA and NEAA; 100EAA = EAA mix, 184.8 g/d, meeting 100% requirement of EAA and 90% requirement of NEAA; 100NEAA = NEAA mix, 158.0 g/d, meeting 90% requirement of EAA and 100% requirement of NEAA; 110NEAA = EAA and double NEAA mix, 501.9 g/d, meeting 100% requirement of EAA and 110% requirement of NEAA; 1MH, 3MH = 1- and 3-methylhisitidine, respectively; ADICP = acid detergent insoluble crude protein; aNDFom = sodium sulfite and alpha amylase–treated NDF corrected for organic matter; AR(1) = first-order autoregressive covariance structure; Asx = Asp and Asn analyzed together; aTTD = apparent total-tract digestibility; BMR = brown-midrib corn silage; CNCPS = Cornell Net Carbohydrate and Protein System; Cov Diet = covariate-period diet; CS = corn silage; DNFA = de novo FA; Exp Diet = experimental-period diet; FA = fatty acid; FTIR = Fourier-transform infrared spectrophotometer; Glx = Glu and Gln analyzed together; LCMS = liquid chromatography–mass spectrometry; MFA = mixed FA; MG = mammary gland; NDFd = digestible NDF; NDICP = neutral detergent insoluble crude protein; NEFA = nonesterified fatty acid; Pd = period; PFFA = preformed FA; PUN = plasma urea N; SBM = soybean meal; TAA = total AA; TFA = total FA; ToD = time of day; Trt = treatment; uNDF240om = sodium sulfite and alpha amylase–treated NDF after 240 h in-vitro rumen fermentation corrected for organic matter; uNDFom = unavailable amylase-treated NDF corrected for organic matter; Z-HILIC = zwitterionic hydrophilic interaction liquid chromatography.

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